Esteban Buz

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Skills

Principal languages: Python (incl. Numpy, Pandas, Scikit-learn, PyTorch), R (incl. tidyverse), SQL, JavaScript, Bash

Analytics/ML: Regression, Bayesian statistics, non-parametric statistics (bootstrapping, permutation), classification (random forest, SVM, XGB), time series/forecasting, LLMs, recommendations

Tools & Frameworks: AWS, Jupyter, R Shiny, Flask, Dash, FastAPI, Docker, Git/Github, LLMs

Domain expertise: data-backed decision making, experimental methods (hypothesis, A/B, and multi-armed bandit), cognition, speech perception/production, linguistics

Experience

Loop Returns Staff machine learning engineer — Remote/Columbus, OH | Jan 2024—Present

- Developed large scale recommendation system handling thousands of merchants, millions of customers and products, and billions of transactions that have meaningfully increased merchant sales and retained revenue
- Developed an ML/LLM-backed image analytics pipeline to automate customer returns and merchant analytics
- Developed an ML/LLM-based exchange system to streamline customer UX and increase retained revenue

SonderMind Senior data/applied scientist — Remote/Denver, CO | Aug 2021—Jan 2024

- Developed and deployed a client-provider recommendation system that increased rate of initial therapy sessions
- Developed an A/B and Multi-armed bandit testing framework for ML experimentation of production services

Qntfy (acquired by SonderMind) Senior data scientist — Remote/Boston, MA | Aug 2020—Aug 2021

- Developed individual and group level mental health measures based on social media data to understand mental health needs at the individual and group level for non-profit organizations who can meet those needs
- Developed analytic and reporting pipelines of large scale social media data (10k> users and millions> of messages) for internal and external stakeholders to understand emergent trends and clusters

Insight Data Science Data science fellow — Boston, MA | Jan 2020—Apr 2020

- Built a tool to predict availability of plants in Massachusetts' public parks with machine learning techniques (e.g. SVM) using millions of plant ID records, climate data, and geo-biome data from the US geological survey

Princeton University Postdoctoral fellow — Princeton, NJ | Sep 2016—Aug 2020

- Developed bootstrapping, permutation, and Bayesian analytic tools in R to quantify asymmetric changes at the lower and upper range of speech outcomes, finding that humans modify their speech at the scale of milliseconds
- Designed experimental protocols to investigate human adaptations to dialects, developed a React, Node.js and PostgreSQL experimental tool to administer these protocols, and an R pipeline to process collected audio data
- Designed and taught an undergraduate seminar on psychological and linguistic approaches to understand conversation, including developing an automatic analysis pipeline in R for students to use in class projects

Vanderbilt University Postdoctoral fellow — Nashville, TN | Sep 2018—Aug 2019

- Modeled speech tone importance for language comprehension using classification techniques (e.g., Bayesian logistic regression) to understand comprehension deficits in individuals with cochlear implants

University of Rochester Graduate researcher and instructor — Rochester, NY | Sep 2010—Aug 2016

- Designed experimental protocols to assess psychological theories of speech change, developed a Flask and PostgreSQL audio/visual web-based recording tool deployed to AWS to administer these protocols, and developed an R analytics pipeline to assess the hypotheses using thousands of speech records
- Contributed to a multi-person effort to design, develop, and deploy a Django+Shiny web-based tool for hundreds of undergraduates to self-design, conduct, and auto-analyze human behavioral studies

Education

PhD in Brain and Cognitive Sciences, University of Rochester — 2016
BA in Cognitive Sciences w/minor in Computer Science, Johns Hopkins University — 2008